PROJECT EXECUTABLE FILES

OptiCrop: Smart Agricultural Production Optimization Engine

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Team ID	SWTID-2026-7253
Team Size	5
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Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Introduction

The Project Executable Files document describes the executable components, configuration files, trained machine learning models, templates, and supporting resources required to successfully run the **OptiCrop: Smart Agricultural Production Optimization Engine**[cite: 30]. These files collectively ensure smooth execution, reliable crop optimization engine predictions, seamless user interaction, and database connectivity[cite: 30].

Executable Components

Executable Component	Description	Purpose
`app.py`	Main Flask application launcher file[cite: 30].	Starts the web application server and coordinates all API endpoint routing loops[cite: 30].
`model.pkl`	Trained Machine Learning core file[cite: 30].	Generates highly accurate agricultural crop compatibility and recommendation predictions[cite: 30].
`encoders.pkl`	Categorical Label Encoder File[cite: 30].	Converts text-based crop and labels features into machine-readable numerical matrices[cite: 30].
`scaler.pkl`	Feature Scaling Configuration File[cite: 30].	Normalizes input soil-climate parameter values before executing optimization engine modeling[cite: 30].
`requirements.txt`	Python Dependency Module Specification[cite: 30].	Installs all required underlying system libraries, core project packages, and framework variants[cite: 30].
`templates/`	User Interface Templates Folder[cite: 30].	Contains the HTML web view pages for input wizards, analytics dashboards, and output panels[cite: 30].
`static/`	Static Script, Styling, and Asset Folder[cite: 30].	Provides dynamic CSS stylesheets, responsive Bootstrap components, client-side JS, and design icons[cite: 30].
`README.md`	Technical System Documentation File[cite: 30].	Outlines the detailed installation commands, application configurations, and manual run procedures[cite: 30].

EXECUTION REQUIREMENTS

Programming Language	Python 3.x[cite: 30]
Web Framework	Flask[cite: 30]
Machine Learning Libraries	• Scikit-learn • Pandas • NumPy[cite: 30]
Database Stack	MySQL[cite: 30]
Development Environment	Visual Studio Code, Jupyter[cite: 30]
Version Control	Git and GitHub[cite: 30]
Deployment Platform	Render with Docker[cite: 30]
Operating System	Windows 11 / Linux[cite: 30]

Key Features of Executable Files

- Highly modularized project directory layout[cite: 30].
- Repeatable application containerization and deployment[cite: 30].
- Secure pipeline loading routines for serialized models[cite: 30].
- Optimized transactional database schema connection mappings[cite: 30].
- Responsive UI optimized for field operator layouts[cite: 30].
- Lightweight and scalable Flask app core microarchitecture[cite: 30].
- Reusable data scaling and formatting component classes[cite: 30].
- Simplified structural expansion paths for additional crops[cite: 30].

PROJECT EXECUTION WORKFLOW

Install Project Dependencies[cite: 30]

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Configure Database Credentials[cite: 30]

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Load Trained Machine Learning Model[cite: 30]

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Launch Flask Core Application[cite: 30]

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Open Targeted Web Browser Endpoint[cite: 30]

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User Secure Account Login[cite: 30]

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Input Granular Field & Soil Parameters

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Execute Multi-Class Machine Learning Prediction

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Display Crop Recommendation & Suitability Result

Conclusion

The executable components of the **OptiCrop: Smart Agricultural Production Optimization Engine** provide a complete, modular, and exceptionally maintainable software solution[cite: 30]. The cohesive grouping of the central Flask application core, pre-calibrated machine learning checkpoints, structured web templates, and localized configuration resources guarantees reliable system execution alongside high-confidence, real-time crop optimization intelligence[cite: 30].